

Stage 3 Outer Race Fault Successfully Replaced — Conveyor Motor A

Asset: AST_MTR_001 (Motor) | Bearing: BRG_001 (SKF6310)
Fault Mode: outer_race_fault | ISO Stage: Stage 3
Learned Case — Knowledge Agent Retrieval Document | DRO Baseline
CASE_001 | Outcome: SUCCESSFUL | Opened: 2026-04-10 | Closed: 2026-04-10

1. Detection Summary

Detection Date	2026-04-10
Detected By	monitoring_agent
Anomaly Score	0.81
vib_rms_mm_s	7.5 mm/s (baseline mean: 2.1 mm/s — z-score: 18σ)
kurtosis	7.2 (FT_001 threshold: 5.0 — exceeded, fault confirmed)
temp_c	81°C (baseline mean: 58°C — rise: 23°C)
bpfo_energy	3.8× (stage_3_vib_multiple: 3.5× — Stage 3 confirmed)
bpfi_energy	0.9× (sub-threshold — inner race not involved)
signal_quality_score	0.99 (clean signal — not a sensor artefact)
NDE Bearing (BRG_002)	Within baseline — fault confirmed as DE-only

2. Agent Diagnosis

Primary Diagnosis	outer_race_fault — Stage 3
Confidence	0.87
Reasoning	kurtosis 7.2 > FT_001 threshold 5.0. bpfo_energy 3.8× > stage_3 multiple 3.5×. Temperature rise 23°C consistent with outer race fault. BPFI sub-threshold.
Differential Considered	inner_race_fault ruled out (bpfi_energy 0.9×, no shaft-frequency modulation).
RUL Estimate	5–9 days from Stage 3 detection (confirmed 7 days in this case)

3. Action Taken

Recommended Action	Urgent bearing replacement — drive end bearing only
Window Selected	CTX_001 — 2026-05-25 02:00 to 06:00 (4 hours, approved, LINE_001)
Parts Required	PART_001 (SKF6310-ZZ, 1 unit, BIN_A1), PART_004 (Synthetic Grease)
Crew Assigned	T. Rodriguez (TECH_001, lead), Raj Kumar (TECH_003, assistant)
LOTO Applied	LOTO_001 — MCC-12, Valve-4 (is_current: true)
Work Order	WO_001 — priority HIGH

4. Findings at Inspection / Teardown

Upon removal of the SKF6310-ZZ bearing from the drive end shaft, outer race showed visible spalling on approximately 35% of the raceway surface, concentrated in a 40mm arc in the loaded zone. Grease was darkened with metallic contamination. Bearing cage was intact. Inner race and rolling elements showed no significant surface damage — fault was localised to the outer race as diagnosed. Shaft journal and housing bore were within tolerance. A

small crack was found in the bearing housing end cover seal — this had allowed fine process dust ingress over time. Seal was replaced as part of this repair.

5. Outcome

Post-Repair vib_rms	1.8 mm/s (QA pass: ≤ 2.5 mm/s) ✓
Post-Repair temp_c	55°C (QA pass: ≤ 60°C) ✓
Kurtosis at 24h	2.1 (QA pass: ≤ 2.8) ✓
BPFO Energy at 24h	0.9× (QA pass: ≤ 1.2×) ✓
QA Result	PASS — all criteria met at 1h and 24h
Actual Duration	7 hours (estimated 6h — extra time for seal replacement)
Actual Cost	INR 18,500
Recommendation Followed	Yes

6. Root Cause Confirmed

Root Cause	contamination_ingress
Contributing Factors	Housing end cover seal degraded (age + thermal cycling). Allowed fine process dust entry over approximately 4–6 months.

7. Lessons Learned

- Outer race fault on SKF6310 in this motor frame consistently shows kurtosis > 5.0 and BPFO > 3.5× at Stage 3. RUL from Stage 3 detection in this case: 7 days.
- Housing seal condition must be inspected at every bearing replacement — replacing only the bearing without checking the seal allows re-contamination.
- NDE bearing (BRG_002) remained healthy throughout — DE and NDE should always be assessed independently.
- Post-repair baseline established: 1.8 mm/s vib, 55°C. Use as expected healthy state for this motor frame.